Eberswalde University for Sustainable Development – HNEE
Department Forest - Wood - Environment

Warsaw University of Life Sciences WULS – SGGW
Faculty of Forestry

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English Thesis Title

Deutscher Titel

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Thesis written under the supervision of

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Eberswalde, 2026

Declaration of the Thesis Supervisor

I hereby declare that this thesis has been prepared under my supervision and I confirm that it meets the conditions for presenting this work in the procedure for the award of a professional title.

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Zusammenfassung

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Keywords - TLS, Grünvolumen, voxel, Semantische Segmentierung, RayCloudTools, Urbaner Waldgarten

Summary

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Summary - Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Keywords - TLS, green volume, voxel, semantic segmentation, RayCloudTools, urban forest garden

Acknowledgements

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List of Acronyms

3DGV	three dimensional green volume
ALS	Airborne Laser Scanning
AOI	area of interest
CSF	Cloth Simulation Filter
C2C	cloud to cloud distance
CV	coefficient of variance
CWD	coarse woody debris
DBH	diameter at breast height
DL	deep learning
eLVV	effective living vegetation volume
FN	false negatives
FP	false positives
FSCT	Forest Structural Complexity Tool
GNSS	global navigation satellite system
GV	green volume
GVA	green volume per area
GVZ	Grünvolumenzahl
LAI	Leaf Area Index
LAD	Leaf Area Density
LiDAR	light detection and ranging
LVV	living vegetation volume
MLS	Mobile Laser Scanning
P2T	Point2Tree
PAI	Plant Area Index
PDAL	Point Data Abstraction Library
RCT	RayCloudTools
TGB	tridimensional green biomass
TLS	Terrestrial Laser Scanning
TP	true positives
tLVV	total living vegetation volume
UAV	unmanned aerial vehicle

1 Introduction

1.1 Background

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1.2 Measuring Green Volume

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1.3 Objectives

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2 Material and Methods

2.1 Study Site

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2.2 Data Acuisition

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2.3 Next Section

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3 Results

3.1 Performance

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3.2 Green Volume Estimates

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4 Discussion

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5 Conclusion

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